Sequel Write-up

Introduction

This challenge revolves around accessing a MySQL database as the root user.

Databases such as MySQL or MariaDB are widely used to store and organize information in a structured, easily accessible manner. The stored data may include usernames, passwords, posts, messages, timestamps, or any other details relevant to the application's purpose.

At the core of every database are tables, which hold information in the form of rows and columns, making it possible to manage and retrieve data efficiently.

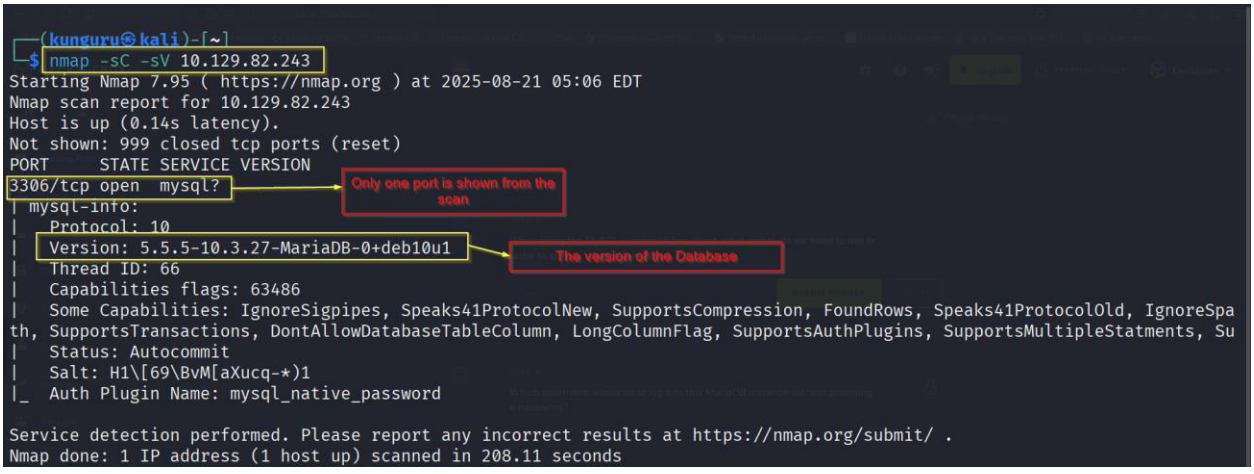
Tools

Nmap

Mysql cli

Kali Linux

Recon



For the initial enumeration, we used Nmap to scan for open ports and the services running on them.

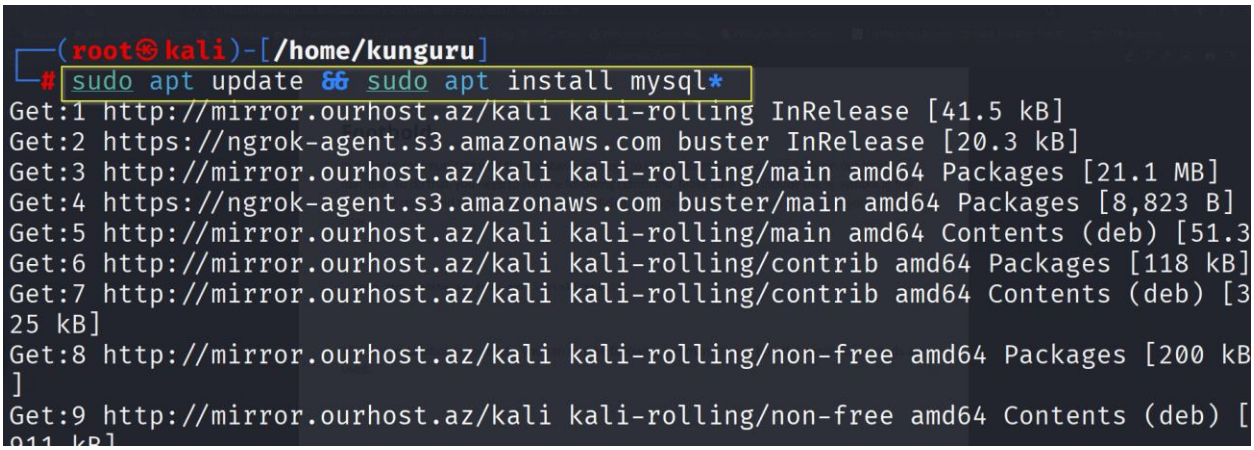
- -sC: runs the default set of Nmap scripts for basic enumeration.
- -sV: enables version detection to identify the exact service versions.

The scan revealed a single open port: 3306, which is running MySQL 5.5.5-10.3.27-MariaDB-0+deb10u1.

MySQL is a database management service used to create, modify, and query databases. It allows interaction with data such as inserting, updating, and retrieving records.

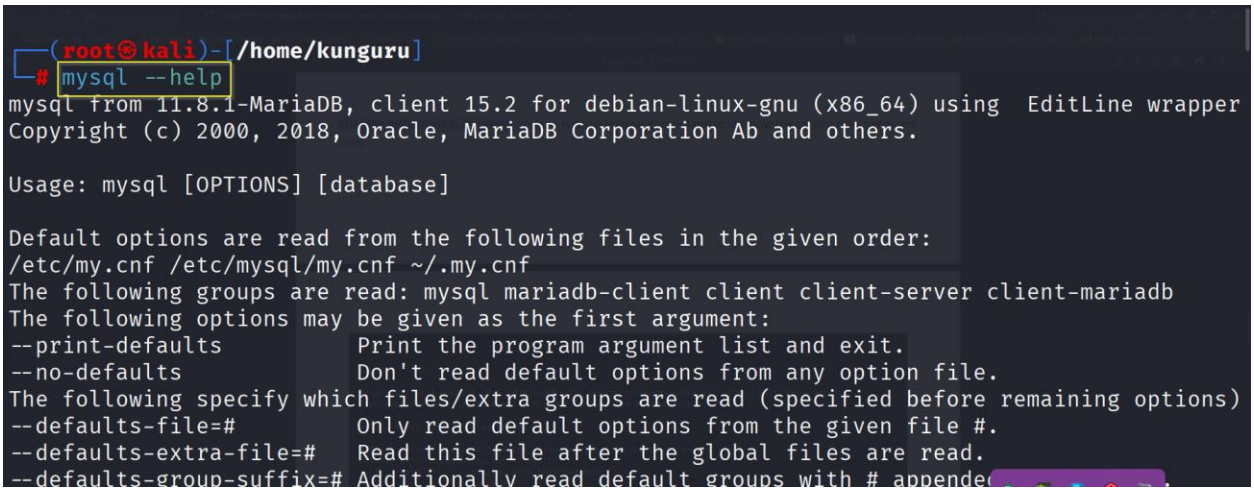
Foothold

To interact with the database, we first need to install the MySQL or MariaDB client on our local machine.



When installing, we append the * wildcard to the package name to ensure that all related MySQL utilities and dependencies are included.

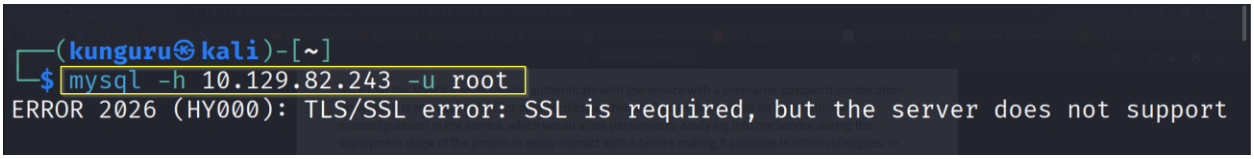
Once the installation is complete, we can use the --help flag to explore available options and familiarize ourselves with the tool's functionality.



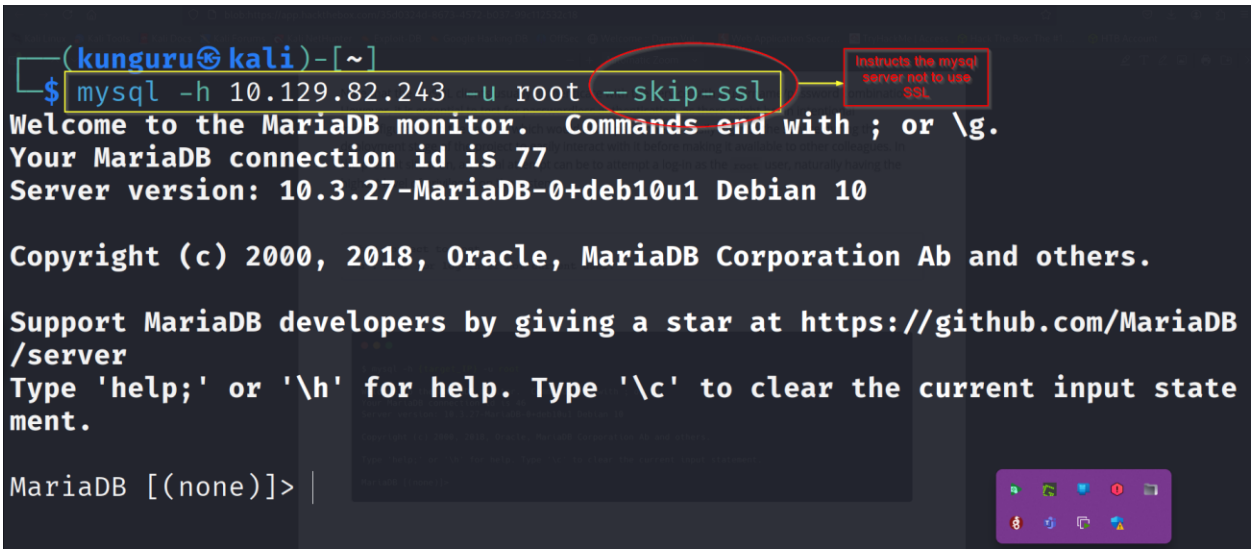
MySQL clients typically authenticate to the service using a username and password combination.

It's important to check for passwordless authentication, as misconfigurations during deployment may leave the service open for easy access.

In this case, the first logical step is to attempt a login as the root user, who by default holds the highest level of privileges on the system.

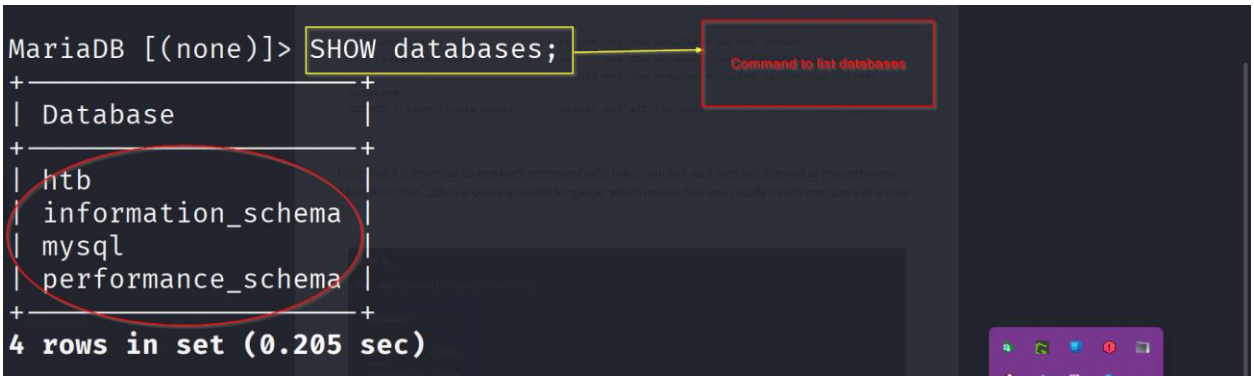


When attempting to connect, we encountered an error indicating that the MySQL client required SSL, but the server did not support it. To resolve this, we disabled SSL in the connection.

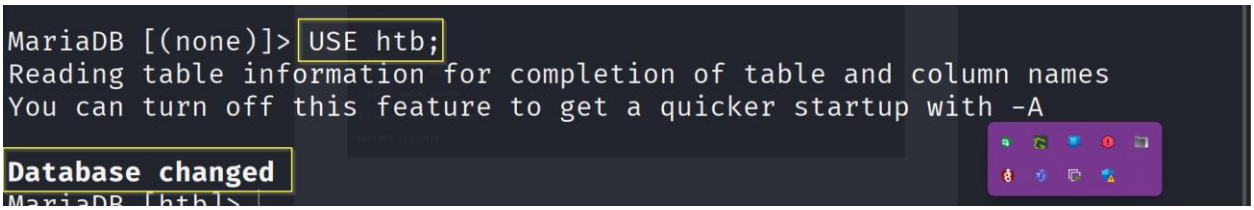


- -h: specifies the target host to connect to.
- -u: defines the username used for login (if different from the current system user).

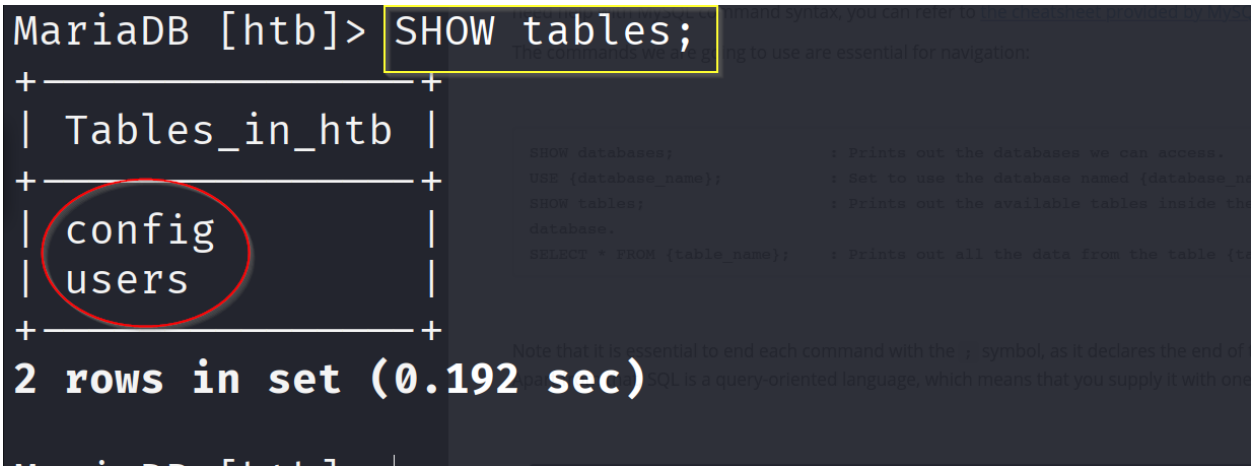
Upon successful connection, we are dropped into the MySQL shell, where we can begin exploring the available databases, tables, and their contents.



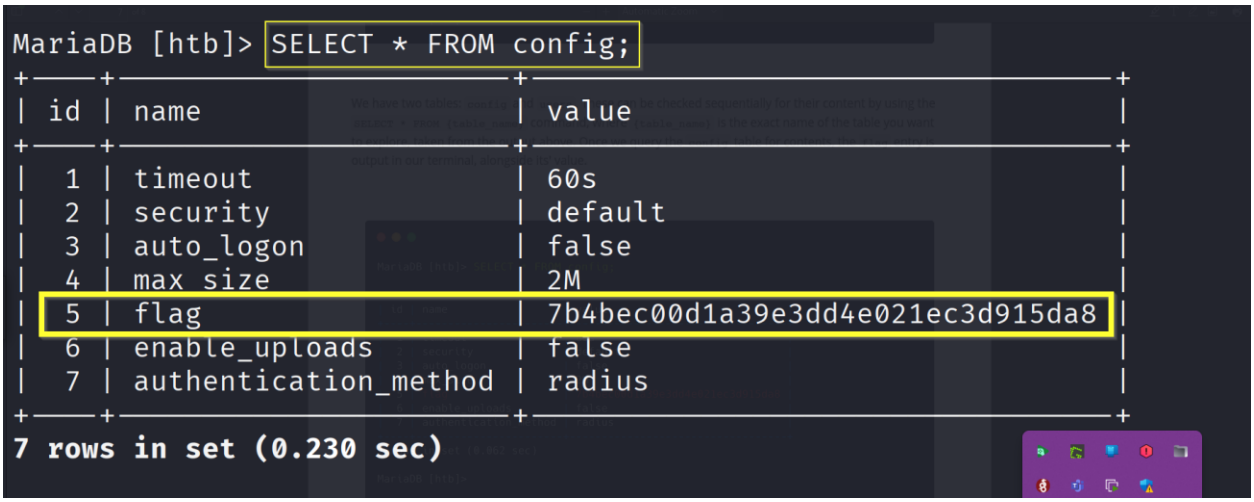
The command `SHOW DATABASES;` lists all accessible databases. From the output, the htb database appears to be the most relevant.



`USE htb;` switches the active database to htb. With the database selected, the next step is to enumerate its tables.



SHOW TABLES; displays all tables within the currently selected database.



`SELECT * FROM {config};` retrieves all records from the config table.

This command allows us to view every entry stored within it, including columns and their respective values.

From the output, we discover the flag: 7b4bec00d1a39e3dd4e021ec3d915da8

This confirms that the config table contained the key information required to complete the challenge.